



College of Sciences
Department of Applied Physics & Astronomy

Course Syllabus

Course Title

Physics 1 Laboratory

Course Code
1430116

Fall Semester, 2026/2027

Course Syllabus

Course Title	Physics 1 laboratory										
Course Code	1430116										
Credit Hours	1				Contact Hours			3			
Course Category		Univ. Req.			Col. Req.		√	Prog. Req.		Others	
Course Type	√		Compulsory						Elective		
Pre-requisite(s)	1430115 - Physics 1										
Co-requisite(s)	1430115 - Physics 1										
Semester	Fall				Year		2026/2027				
Instructor Name	AbdulRaheem Khudada Mohammed										
Office Location	M7-005A (A11-005A)										
Tel. No.	(06-5050296) or Ext. 2296										
Email	aalzubaidy@shariah.ac.ae										
Lecture Times	Sec (07L)		Mon. (8:00-10:40)						M12-106		
	Sec (09L)		Thur. (8:00-10:40)						M12-106		
	Sec (10L)		Wed. (8:00-10:40						M12-106		
	Sec (11L)		Wed. (11:00-13:40)						M12-106		
	Sec (31L)		Tue. (11:00-13:40)						M12-106		
	Sec (39L)		Thur. (11:00-13:40)						M12-106		
Office Hours	Mon. (11:00-12:00), M7-005A Thur. (14:00-15:00), M7-005A										

Course Description (as in the catalogue):

This course includes 11 experiments covering the following topics: General Instructions, Analysis and Graphing Data, Measuring devices and density, Vectors, Free Fall, Uniformly Accelerated Motion, Projectile Motion, Newton's 2nd Law, Friction, Conservation of Mechanical Energy, Conservation of Linear Momentum and Ballistic Pendulum.

Course Objectives/Goals(optional):

Teaching Methods:

During the laboratory sessions students implement the knowledge gained from the theory. The main ideas of the experiment are explained and discussed at the beginning of the lab using a PowerPoint presentation. Students are then introduced to the method of conducting the experiment and the proper way to use devices.

All documents, teaching material, lab reports, and announcements are continuously uploaded within the Blackboard system.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. Describe mechanics and thermodynamics phenomena and explain the concepts and principles behind the experiment.
2. Apply experimental techniques and skills in error estimation, error propagation, and graphical analysis.
3. Use instruments (e.g., micrometer, Vernier caliper, combi-light barriers...etc.) and computer-based technology in the collection, analysis, and interpretation of data.
4. Function effectively in a team and demonstrate effective writing skills through written laboratory reports.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
1- Identify, formulate, and solve broadly defined technical or scientific problems by applying knowledge of mathematics and science and/or technical topics to areas relevant to the discipline.	- Describe mechanics and thermodynamics phenomena and explain the concepts and principles behind the experiment.
2- Formulate or design a system, process, procedure, or program to meet desired needs.3	- Apply experimental techniques and skills in error estimation, error propagation, and graphical analysis.
3- Develop and conduct experiments or test hypotheses, analyze, and interpret data and use scientific judgment to draw conclusions.2	- Use instruments (e.g., micrometer, Vernier caliper, combi-light barriers...etc.) and computer-based technology in the collection, analysis, and interpretation of data.
4- Integrate the impact of technical and/or scientific physics solutions in a global, economic, environmental, and societal context.	- Describe mechanics and thermodynamics phenomena and explain the concepts and principles behind the experiment.
5- Communicate effectively with a range of audiences.	- Function effectively in a team and demonstrate effective writing skills through written laboratory reports.
6- Determine ethical and professional responsibilities related to Physics applications.	- Apply experimental techniques and skills in error estimation, error propagation, and graphical analysis.
7- Function effectively on teams that establish goals, plan tasks, meet deadlines, and analyze risk and uncertainty.	- Function effectively in a team and demonstrate effective writing skills through written laboratory reports.

Weekly Distribution of Course Topics/Contents

Week	date	Exp.	Application	comments	Course SLO
1 st	24-27 Aug. 2026	Safety and General Instructions: (purpose of the lab, lab manual, grading, lab preparation, students' groups, lab procedure, lab reports).	Following the correct safety rules.	Following the correct safety rules. Link for the Online Safety Training System.	
2 nd	31 Aug -3 Sep. 2026	Exp (1): Error calculation, Analyzing, and graphing	Machine accuracy and an acceptable range of electronic devices.	PP presentation recorded lecture & video, online session using Bb. Collaborate. Link to manual.	2,4
3 rd	7-10 Sep. 2026	Exp (2): Measuring devices and density	Floating and Sinking, Ships and Submarines, Oil Spills, Plumbing Systems, Airplane Weight Distribution.	Link to manual. Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources	2,3,4
4 th	14-17 Sep. 2026	Exp (3): Free Fall	Dams and Hydroelectric Power, Spacecraft in Orbit, Sky Diving, Shells Falling after Firing	Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources Link to manual.	1,3,4
5 th	21-24 Sep. 2026	Exp (4): Vectors	Flying, GPS, Balancing objects. Avoiding crosswind and military usage.	Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources Link to manual.	4
6 th	28 Sep-1 Oct. 2026	Exp (5): Projectile Motion	Firing a canon, Basketball, Javelin throw, Archery, Golf ball, Gun fire.	Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources	1,3,4
7 th	5-8 Oct. 2026	Exp (6): Newton's Second Law	Pushing car, trucks, shopping cart. Rocket launch, Hitting a ball, Car crash, Racing car.	Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources	1,2,4
8 th	12-15 Oct. 2026	Midterms (Theory) and quizzes			
9 th	19-22 Oct. 2026	Exp (7): Friction	Walking, Driving, Writing, Skating, meteorites combustion, Applying brakes.	Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources	1,2,4
10 th	26-29 Oct. 2026	Exp (8): Work-Energy Theorem	Braking Systems, Engine Efficiency, Car Safety, Roller Coasters, Skiing/Sledding, Mechanical Presses,	Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources	1,2,4

Week	date	Exp.	Application	comments	Course SLO
			Hydraulic Systems, Lifting Objects		
11 th	2-5 Nov. 2026	Exp (9): Conservation of Mechanical Energy	Wrecking ball, hammer, Windmill, Electric motors, Hydropower plants, Bow and an arrow.	Link to manual. Link to online exp. Presentation using Bb Collaborate Ultra Extra links and	1,3,4
12 th	9-12 Nov. 2026	Exp (10): Conservation of Linear Momentum	Gun and Bullet Mechanism Collision of Two Cars Launching of Rockets Firecrackers Billiards and Snooker Ice Skater.	Link to manual. Link to online exp. Presentation using Bb Collaborate Ultra Extra links and resources	1,2,4
13 th	16-19 Nov. 2026	Missing labs & Revision (Only one makeup experiment is allowed with an acceptable excuse)			
14 th	23-26 Nov. 2026	Final exam			

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Course Work: [4 Quizzes 30 %]	During the course scheduled hours	30
Experiments and Laboratory report	Every week	30
Final Examination	It will be announced later	40
Total		100

** You can modify/add other tools relevant to the course.

Assessment tools

Quizzes:

4 quizzes are scheduled per semester, each quiz typically covering two experiments.

The top 3 scores will be counted.

Lab Reports:

9 lab reports during the semester.

Reports for experiments that the student did not attend will not be counted, even if the student participated in preparing or writing the report.

The instructor will divide students into groups; each group consists of 2-3 students that can work together as a team. Weekly lab reports will be completed after performing the related experiment and are due before the date of the next experiment.

Exams:

Final exam is scheduled for this lab during the last week of study.

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UOS Competency
Develop a greater conceptual understanding of mechanics and thermodynamics phenomena and explain the concepts and principles behind the experiment.	- PP presentations - problem solving, - Questioning and answering. - Practical work - Teamwork and Discussion	Lab reports Quizzes Exams	70 % of students get at least 70 % of the allowed points.	- Critical and analytical thinking - Oral communication - Intellectual curiosity and lifelong learning - Problem-solving - Numeracy - Innovation / creative thinking
Develop good experimental techniques and skills in error estimation, error propagation, and graphical analysis.	- PP presentations - problem-solving, - Questioning and answering. - Practical work - Teamwork and Discussion	Lab reports Quizzes Exams	70 % of students getting at least 70 % of allowed points.	- Critical and analytical thinking - Oral communication - Intellectual curiosity and lifelong learning - Problem-solving - Numeracy
Use instruments (e.g., micrometer, combi-light barriers, sensors, etc.) and computer-based technology in the collection, analysis, and interpretation of data.	- PP presentations - problem-solving, - Questioning and answering. - Practical work - Teamwork and Discussion	Lab reports Quizzes Exams	70 % of students getting at least 70 % of allowed points.	- Critical and analytical thinking - Intellectual curiosity and lifelong learning - Problem-solving - Numeracy - Innovation / creative thinking
Demonstrate effective writing skills through written laboratory reports.	- problem-solving, - Questioning and answering. - Practical work - Teamwork and Discussion	Lab reports Quizzes Exams	75% of students getting at least 80 % of allowed points.	-Written communication - Teamwork/collaboration - Oral communication - Innovation/creative thinking - Problem-solving - Autonomy and resiliency - Work ethics/ professionalism

Teaching and Learning Resources:

Reference Books:

Textbook: Laboratory Experiments in Physics 1 and the online lab report.

(All are available in the Blackboard system)

Other References:

- PowerPoint presentations for all experiments prepared by instructors and contain objectives of the experiment, definitions, theoretical background, procedures, analysis, safety rules, and some real-colored photos from the lab, all are available within the blackboard system
- Principles of Physics, 10th Edition International Student Version, David Halliday, Robert Resnick, Jearl Walker, ISBN: 9781118230749 ©2014 Wiley
- **Physics for Scientists and Engineers 9th edition.** Raymond A. Serway; John W. Jewett. ISBN 13: 9781133947271, ISBN: 1133947271© 2013 Brooks/Cole

Supplementary Material:

Lab Coat, Laptop, Lab report of the exp., Graph papers, Scientific Calculator, Pencil, Eraser, Protractor, and Ruler.

Attendance policy:

Attendance is compulsory. A student missing 10% (2 labs) of the total allocated course hours will receive a 1st warning notice and a student missing 15% (3 labs) will receive a 2nd warning notice. A student missing more than 20% (4 labs including the introduction lab) will be forced to withdraw (in accordance with the university regulations).

Attendance:

10%	two Labs (1 st Warning)
15%	three Labs (2 nd Warning)
20%	four Labs (FORFIET) FA

Important dates

Sr.	Date	Subject
1.	24/8/2026	Classes begin
2.	27/8/2026	Last day for Add/Drop
3	(12-22)/10/2026	Midterm exams (Theoretical courses)
4.	29/10/2026	Last day for dropping courses without 'F'
5.	19/11/2026	Last day for withdrawal from the semester
6	(23-26)/11/2026	Final Exams for Labs
7.	3/12/2026	Classes end
8	(5-15)/12/2026	Final Exam (Theoretical courses)

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about the general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Weekly lab reports:

Weekly lab reports will be completed after performing the related experiment. The weekly lab reports are due before the date of the next experiment. The lab report follows the provided online form.

Group Report Rules:

- I- All members in the group must contribute equally when working on the experiment as well as writing the lab report.
- II- It is the responsibility of each group member to honestly describe the work that they did during the experiment and in writing the report.
 - No one should take advantage of the other group members.
 - Report to the course instructor or your TA if you feel that other group members are taking advantage of your work.
- III- Each member in the group must understand the concepts behind the entire experiment regardless of which portions of the report or experiment a person is responsible for. This is especially important for the exams.

Communication Plan:

E-mail is the primary communication method in this course. I will respond to students' e-mails within 24 hours during the weekdays.

Common grading rubrics

Rubric for Grading Lab Reports in Physics 1 Lab

Performance elements	Maximum Mark 10				Instructor Evaluation
Title and Objectives (1)	All elements are thoroughly presented; extremely well organized. (1)	All elements present with occasional disorganization. (0.5)	Some elements missing. (0.5)	Elements missing; minimal. (0)	
Results & Analysis (5)	The data: make sense, appropriate significant figures, suitably tabulated, correctly analyzed, and accompanied by uncertainty analysis. (5)	Minor parts are missing. (4)	Some parts are missing. (3)	Inadequate data & analysis. (1)	
Post-Lab Questions (2)	All correct. (2)	Mostly correct. (1.5)	50% correct. (1)	Mostly wrong. (0)	
Conclusions (2)	Appropriately addressed the	Minor parts are missing.	Some parts are missing.	Inadequate conclusion.	

	objective, results & analysis with stated final uncertainty. Significant sources of uncertainty are adequate. (2)	(1.5)	(1)	(0)	
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The table summarizes the alignment between Physics 1 laboratory experiments and the theoretical topics covered in Physics 1.

11 experiments in mechanics, covering the topics in Physics 1 course as shown below.

Week	Physics (1) (1430115)	Physics (1) Lab (1430116)
1.	Introduction: All Chapters	Exp (1): Introduction & Error calculation.
2.	Ch (1): Measurement	Exp (2): Measuring devices and density.
3.	Ch (2): Motion Along a Straight Line	Exp (3): Free Fall
4.	Ch (3): Vectors	Exp (4): Vectors
5.	Ch (4): Motion in Two and Three Dimensions	Exp (5): Projectile Motion
6.	Ch (5): Force and Motion-I	Exp (6): Newton's Second Law
7.	Ch (6): Force and Motion-II	Exp (7): Friction
8.	Ch (7): Kinetic Energy and Work	Exp (8): Work-Kinetic Energy Theorem
9.	Ch (8): Potential Energy and Conservation of Energy	Exp (9): Conservation of Mechanical Energy
10.	Ch (9): Center of Mass and Linear Momentum	Exp (10): Conservation of Linear Momentum
11.	Ch (10): Rotation	Exp (11): Rotational Dynamics (Standby) or Uniform Circular Motion (Standby)